Systematic Feasibility Assessment of Geomechanical Risks for CO_2 Injection and Long-Term Storage Incorporating Rock Properties Degradation and Thermal Effects

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ABSTRACT: As part of carbon capture and storage feasibility assessment, geomechanical risk analysis is conducted in field of interest to determine maximum injection pressure and associated geomechanical risks during injection and long-term storage. The objective of the study is to evaluate the feasibility of injecting and storing CO_2 in the field by assessing CO_2 containment and leakage risks, in particular "showstoppers", for decision-making on progressing to comprehensive coupled geomechanical study and mature the field as a CO_2 storage site. Using the rock mechanical properties from calibrated 1D geomechanical models as input, caprock integrity analysis is conducted and maximum injection pressure is determined to ensure the caprock does not experience tensile and shear failure during CO_2 injection. If there are faults penetrating the caprock and reservoir of interest, fault stability analysis is conducted on the faults to ensure they remain stable with reduction in maximum injection pressure, if required. Rock properties degradation due to CO_2-rock interaction and thermal effects on geomechanical containment risks are incorporated by using rock properties degradation and thermal properties database of the area and estimated temperature changes in the reservoir and caprock. Upon finalization of the maximum injection pressure, reservoir expansion and surface uplift are conducted to determine the maximum uplift. The methodology has been applied to one of the CO_2 storage candidates located in offshore Peninsular Malaysia. Changes of minimum horizontal stress at caprock during reservoir pressure increase were calculated to determine the injection pressure that will cause tensile or shear failure. Fault stability analyses were conducted on all the faults intersecting the formations of interest by using the rock strength and stress properties from the nearest well to the faults, and the faults were found to remain stable. Finally, reservoir expansion for each of the formations of interest was calculated and used as part of the input for seabed uplift analysis. Low seabed uplift was determined at the end of injection and will not affect the surface facility integrity.

1. INTRODUCTION

One of the best alternative methods of produced carbon dioxide (CO_2) disposal to meet net zero carbon emission is CO_2 capture and geological storage. With more countries committed to the reduction of greenhouse gas emission following their ratification of the Paris Climate Agreement, there has been increasing global interest in the geological storage (Baklid et al., 1996, Bissell et al., 2011, Mustafa et al., 2021 and Tewari et al., 2022). However, there are numerous geomechanical challenges and risks associated with CO_2 injection and storage in depleted reservoirs, saline aquifers and dry geological structures that are required to be addressed. The key challenges include maintaining the seal integrity throughout the injection operation and storage life. The pressure and temperature changes will give rise to changes in stresses and pose risks to fault re-activation and breach of caprock integrity. These processes and mechanisms can be complex and need to be modelled simultaneously as they inter-relate between them (Tan et al., 2022 and Chidambaram et al., 2023). A comprehensive assessment of the geomechanical-related risks to manage CO_2 containment and mitigate leakage risks requires a coupled geomechanical study (Masoudi et al., 2011, Masoudi et al., 2013, Chidambaram et al., 2021 and Mustafa et al., 2022). However, it is resource intensive and time-consuming which are not warranted for preliminary feasibility assessment of potential CO_2 geological storage site. An analytical approach based on 1D geomechanical models may be implemented instead which is solely for preliminary assessement of the risks. The goal is to identify any "showstoppers" from geomechanical perespective that will make the geological structure to be not viable as CO_2 storage site (Musa at al., 2023 and Tan et al., 2024).

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This paper describes the geomechanical processes, mechanisms and the analytical approach for preliminary assessment and mitigation of CO_2 leakage risks associated with CO_2 injection for geological storage. The illustration of the analytical approach and results are based on a preliminary study conducted on Field X which has been selected as a potential CO_2 storage site in offshore Peninsular Malaysia. The objective of the study is to evaluate the feasibility of injecting and storing CO_2 in the field by assessing CO_2 containment and leakage risks, in particular "showstoppers", for decision-making on progressing to comprehensive coupled geomechanical study and mature the field as CO_2 storage site.

2. GEOMECHANICAL PROCESSES AND MECHANISMS

CO_2 injection and storage operation will result in fluid pressure increase, temperature change due to cooling by the injected CO_2 and chemical interaction between CO_2-rich brine and the rocks. For the case of injecting and storing in a depleted reservoir, the reservoir will undergo pressure depletion and associated porosity reduction and compaction prior to the injection which will be transmitted to the surface in the form of subsidence. These changes and interaction will impact the stress condition both in the reservoir and the overburden (Masoudi et al., 2011, Masoudi et al., 2013 and Mustafa et al., 2021).

An increase in fluid pressure of the reservoir will generate increased total horizontal stresses and vice-versa for reservoir depletion (Fjaer et al., 2008 and Zoback, 2010). With the increase in pressure of the fluid which is partially supporting the stresses acting on the reservoir rock, the rock wants to expand but it is unable to expand laterally. Hence, there will be an increase in the total horizontal stresses to compensate for the tendency of the rock wanting to expand. As the total horizontal stress change is smaller than the fluid pressure change, the change in the effective horizontal stresses is reverse i.e., increase in the total horizonal stress will be associated with a decrease in the effective horizontal stress. In the case of cooling the reservoir rock, the rock has tendency to contract and hence, there will be a reduction in total horizontal stresses and increase in effective horizontal stresses. The thermal effect is much more pronounced in the reservoir where it is dominated by convective heat flow, whilst it is a much slower process in the caprock with conductive heat flow due to its low permeability. The total horizontal stress change in the reservoir will cause reverse stress change of smaller magnitude in the immediate caprock e.g., total horizontal stress increase in reservoir during injection will give rise to total horizontal stress decrease in the caprock to maintain stress equilibrium (Fig. 1). In addition, the pressure and stress changes, and the CO_2-rich brine-rock interaction will alter the mechanical properties as well as petrophysical properties of the rock. Furthermore, even if a membrane seal has been generated in a fault through deformation processes, the seal can be breached if the fault is re-activated during the CO_2 injection. The risk of fault re-activation is largely dependent on various fault and rock properties, pore pressure, temperature, relative stress magnitudes and generation of shear stress along the fault, and how the in-situ stresses will evolve under injection condition and operation.

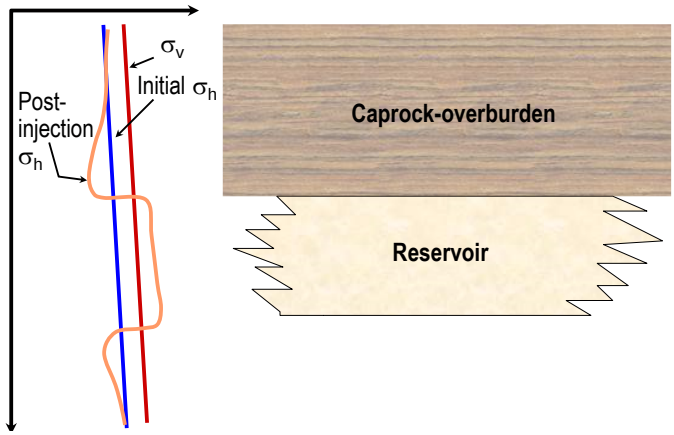


Fig. 1. Illustration of initial and post-injection total minimum horizontal stress (σ_h) and vertical stress (σ_v) profiles in reservoir and caprock.

Assessment is required on the impact of the complex inter-related changes in the stresses, pressure and temperature, associated deformation on cap rock, fault stability and seal integrity, compaction and subsidence to manage CO_2 containment and mitigate risks related to leakage. The assessment of the geomechanical-related challenges and risks require delineation of the vertical and horizontal stresses, pore pressure and the other geomechanical parameters as well as rock mechanics and petrophysical properties of the field.

3. WORKFLOW FOR FEASIBILITY ASSESSMENT OF GEOMECHANICAL CONTAINMENT RISKS

The workflow for the feasibility assessment of geomechanical containment risks is shown in Fig. 2 and the steps are described below.

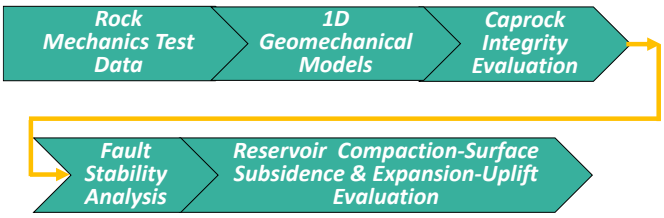


Fig. 2. Feasibility assessment workflow for geomechanical containment risks.

3.1. Data Gathering

Prior geomechanical study and data availability assessment is conducted to determine if there are any rock mechanics test data and 1D or 3D geomechanical models have been constructed in the field. This will be the main reference for the feasibility assessment and to determine whether there is any need for 1D geomechanical model construction. However, the available 1D geomechanical model needs to be critically reviewed and validated prior to adopting it for the feasibility assessment.

In the absence of 1D or 3D geomechanical model in the field, the model for at least one offset well needs to be constructed and validated. A critical well that penetrated the formations of interest is selected and quality assurance and quality check (QAQC) need to be conducted to ensure that the well contained

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the necessary well logs data such as gamma ray, density and compressional and shear sonic logs. In addition, key downhole test and measurement data such as good quality leak-off test and formation pressure measurement data are required. In the case where only compressional sonic log is available, synthetic shear sonic log needs to be generated and cross-checked based on robust methodologies.

3.2. 1D Geomechanical Model

The geomechanical model contains elastic and strength properties (Young's modulus, Poisson's ratio, unconfined compressive strength, angle of internal friction and tensile strength), pore pressure, and in-situ principal stress magnitudes and orientation (vertical or overburden stress, and minimum and maximum horizontal stresses). The predicted rock properties should be calibrated with available measured elastic and strength properties from rock mechanics tests including either single or multi-stage triaxial tests conducted at a range of effective confining stresses including in-situ effective confining stress, and Brazilian tensile test. Pore pressure can be predicted using compressional sonic and/or resistivity logs and should be calibrated with available measured formation pressure data.

Vertical or overburden stress can be calculated with density log whilst the minimum and maximum horizontal stresses can be estimated with poroelastic horizontal strain model (Fjaer et al., 2008). The minimum horizontal stress should be calibrated with good quality leak-off test data or preferably extended leak-off test data. Pressure vs time/volume pumped plots should be examined to ascertain whether the reported data corresponds to leak-off pressure, breakdown pressure or propagation pressure.

Finally, wellbore stability analysis is conducted to validate the geomechanical model by history-match the predicted formation failures, both shear and tensile failures, against available caliper and image logs, and recorded drilling experiences and observations.

3.3. Rock Properties Degradation

The evaluation of changes in rock mechanical and petrophysical properties due to CO_2 interaction is based on data obtained from CO_2-rock interaction tests. Rock mechanics tests including triaxial test, tensile strength test and uniaxial pore volume compressibility test are conducted on pre- and post-CO_2 treated "sister" core samples of reservoir and caprock to determine the changes in the rock properties due to CO_2-rock interaction (Masoudi et al., 2013). It is found from comprehensive CO_2 interaction laboratory studies that the interaction resulted in reduction in rock strength properties and Young's modulus by up to 26% whilst the increase in Poisson's ratio is up to 10%. The increase in Poisson's ratio with reduction in rock stiffness (Young's modulus) is typical for rock behaviour. The properties degradation database is applied to the properties of the rock materials in sensitivty analysis, and similar changes in Young's modulus and strength properties due to CO_2-rock interaction may also be applied to the fault stiffness and strength properties, respectively.

3.4. Thermal Effects

The key CO_2 cooling impacts on caprock integrity and reservoir rock behaviour associated with reduction in total horizontal stresses are:

- Hydraulic fracture initiation at injector due to cooling and fracture propagation under reduced fracture gradient in reservoir and through caprock
- Fault re-activation due to stress changes in reservoir and overburden (caprock) faults associated with cooling
- Shear failure in caprock due to stress changes and stress transfer between reservoir and caprock

The temperature change is localized in the vicinity of the CO_2 injectors and typically within 750 m lateral distance from the injectors at the end of injection. At the start of the CO_2 injection, the CO_2 plume initially moved towards the top of the reservoir due to buoyancy effect. However, as the injection continued, the CO_2 plume is pushed downward deeper into the reservoir due to higher density of the CO_2-rich brine or supercritical CO_2. It is worth noting the magnitude of the total horizontal stress is also impacted by the pressure increase as the injection progresses whereby pressure increase will result in horizontal stress increase.

3.5. Caprock Integrity Evaluation

The caprocks of the identified reservoirs of interest within the wells are analysed for their integrity during the CO_2 injection. The changes in horizontal stresses in the reservoirs and associated stress changes in the caprocks due to stress transfer are calculated. Firstly, maximum safe injection pressure is determined with applied safety factor that will avoid tensile failure. This is followed by evaluating whether shear failure in the caprock will be induced by the maximum injection pressure with incorporation of uncertainty analysis on the strength properties. The maximum injection pressure should be reduced if shear failure is induced in the caprock.

3.6. Fault Stability Analysis

Following the caprock integrity evaluation, the next step is fault stability analysis. The normal and shear stresses acting on the fault planes are calculated for their corresponding fault dips and azimuths before and after CO_2 injection. Shear failure analysis is conducted based on the calculated normal and shear stresses and if it is found that any of the faults is unstable at end of CO_2 injection, the maximum injection pressure should be reduced. Uncertainty analysis needs to be conducted on the fault dip, azimuth and strength properties.

3.7. Reservoir Compaction/Expansion-Seabed Subsidence/Uplift Evaluation

Reservoir compaction at the end of production for brown field and/or reservoir expansion at maximum injection pressure is determined together with rock properties from the 1D geomechanical models and effective reservoir thickness. The corresponding surface subsidence and/or uplift are calculated based on the predicted reservoir compaction/expansion together with overburden elastic properties and reservoir geometries. The surface subsidence or uplift values are used for surface structure integrity evaluation and design.

The application of the feasibility assessment workflow for geomechanical containment risks is illustrated via preliminary study on a CO_2 storage project in Field X in offshore Peninsular Malaysia.

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4. OVERVIEW OF GEOMECHANICAL STUDY ON CO_2 STORAGE PROJECT IN FIELD X

As geomechanical models are not available for the field, data assessment was conducted to determine the availability of the necessary well logs data and key downhole test and measurement data. In addition, daily drilling reports and end of well reports were reviewed to extract geomechanics-related drilling events and observations in the wells.

1D geomechanical models were constructed for selected offset wells which involved determination of rock elastic and strength properties, calculation of overburden stress from density log, prediction of pore pressure, determination of stress direction and computation of in-situ stress magnitudes. The predicted pore pressure and minimum horizontal stress are calibrated with measured formation pressure data and leak-off test data, respectively. Wellbore stability analysis was conducted with the 1D geomechanical models and validated against caliper log and the extracted geomechanics-related drilling events and observations.

The data contained in the models were extracted and used in the caprock integrity analysis, fault stability evaluation, and compaction-subsidence and expansion-uplift analyses. The key outputs are the recommended CO_2 maximum injection pressure limit, potential of fault re-activation, seabed subsidence at end of production and seabed uplift at end of injection.

5. 1D GEOMECHANICAL MODEL CONSTRUCTION

A total of eight offset wells were selected for 1D geomechanical model construction. The selection criteria were mainly based on data availability, especially sonic log, spatial coverage of the field by the well location and transection of the reservoirs of interest for the potential CO_2 storage site. The data audit conducted is summarized in Table 1. As only compressional sonic log is available in all the selected offset wells, synthetic shear sonic log was generated based on petrophysics principles and information from adjacent fields.

The construction of the 1D geomechanical model started with the calculation of the overburden stress. The method adopted for the overburden stress calculations incorporated synthetic density extrapolation based on power law equation from top of bulk density log to the seabed. After the overburden stress has been calculated, pore pressure prediction was conducted using Eaton method with compressional sonic and resistivity logs, and the overburden stress used as the total stress in the pore pressure prediction calculation. The pore pressure prediction is calibrated with measured formation data, and kick and connection gas information.

The horizontal stress directions were determined based on breakout analysis using oriented caliper log and cross-checked with stress maps in Tjia et al., 2010 and Tingay et al., 2012 for overall consistency.

Since there are no available measured rock properties from rock mechanics tests, the rock elastic and strength properties were generated solely using log data based on rock property correlations including the conversion of dynamic to static elastic properties. The main logs utilized with the correlations are gamma ray, compressional sonic, synthetic shear sonic and density logs.

Table 1. Data audit conducted on offset wells.

WELL/FIELD DATA	A ✓ ✗	B ✓ ✗	C ✓ ✗	D ✓ ✗
Year drilled	1969	1985	1985	2006
1D Mechanical Earth Model (mechanical rock properties + stresses)	✗	✗	✗	✗
Formation Evaluation & Wells		-	-	-
Wireline and LWD Logs:		-	-	-
Gamma Ray	✓	✓	✓	✓
Density	✓	✓	✓	✓
Sonic DTC	✓	✓	✓	✓
Sonic DTS	✗	✗	✗	✗
Caliper	✓	✓	✓	✓
Neutron	✓	✓	✓	✓
Well Survey (Trajectory)	✓	✓	✓	✓
Casing Scheme, completion diagram	✓	✓	✓	✓
Pressure data (MDT/RFT/stethoscope)*	✓	✓	✓	✓
Formation tops*	✓	✓	✓	✓
Wellbore images (OBMI, UBI)	✗	✗	✗	✗
Drilling Data				
FIT/LOT/Mini-frac Tests Analyses	✓	✓	✓	✓
Mudweight MW/ Equivalent circulating density ECD	✓	✓	✓	✓
Daily Drilling Reports	✗	✗	✗	✗
Drilling Summaries	✓	✓	✓	✓
Prior Wellbore Stability (WBS) Reports	✗	✗	✗	✗
Prior Geomechanics study Reports	✓	✓	✗	✓
End Of Well Report	✓	✓	✓	✓

WELL/FIELD DATA	E ✓ ✗	F ✓ ✗	G ✓ ✗	H ✓ ✗
Year drilled	1991	1991	1992	1992
1D Mechanical Earth Model (mechanical rock properties + stresses)	✗	✗	✗	✗
Formation Evaluation & Wells				
Wireline and LWD Logs:				
Gamma Ray	✓	✓	✓	✓
Density	✓	✓	✓	✓
Sonic DTC	✓	✓	✓	✓
Sonic DTS	✗	✗	✗	✗
Caliper	✓	✓	✓	✓
Neutron	✓	✓	✓	✓
Well Survey (Trajectory)	✓	✓	✓	✓
Casing Scheme, completion diagram	✓	✓	✓	✓
Pressure data (MDT/RFT/stethoscope)*	✗	✗	✗	✗
Formation tops*	✓	✓	✓	✓
Wellbore images (OBMI, UBI)	✗	✗	✗	✗
Drilling Data				
FIT/LOT/Mini-frac Tests Analyses	✓	✓	✓	✓
Mudweight MW/ Equivalent circulating density ECD	✓	✓	✓	✓
Daily Drilling Reports	✗	✗	✗	✗
Drilling Summaries	✓	✓	✓	✓
Prior Wellbore Stability (WBS) Reports	✗	✗	✗	✗
Prior Geomechanics study Reports	✗	✗	✗	✗
End Of Well Report	✓	✓	✓	✓

Following generation of the rock properties, the horizontal stresses were determined using the poroelastic horizontal strain model. The horizontal stress magnitude profiles were generated using the computed rock elastic properties contained in the 1D geomechanical models as input. As there is no extended leak-off test (XLOT) data in the field, only leak-off test (LOT) and formation integrity test (FIT) data were used to calibrate/constrain the calculated minimum horizontal stress. The values of both maximum and minimum horizontal strains are refined to overall match the measured leak-of test and formation integrity test data in the wells.

Finally, using the computed rock properties, pore pressure and in-situ stresses of the model, wellbore stability analysis is conducted to ensure the predictions overall matched with the caliper logs and recorded geomechanics-related drilling experiences and observations. Fig. 3 shows the wellbore stability analysis predictions for one of the offset wells. The actual mud weight information is extracted from drilling reports for the well, while the equivalent circulating density (ECD) is assumed to be 0.5 ppg above the static mud weight. The synthetic failure image from the wellbore stability analysis is compared against the trend of hole enlargement based on

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caliper log. The wellbore stability analysis predictions showed good overall consistency with the hole enlargements from caliper logs and the recorded drilling experiences and observations.

6. GEOMECHANICAL RISK ANALYSIS AND RESULTS

6.1. Maximum Injection Pressure and Caprock Integrity Analyses

The rock properties and stresses contained in the 1D geomechanical models are used as input for caprock integrity analysis. The objective of the analysis is to ensure the injected CO_2 remains within the reservoirs throughout the injection phase and long-term storage by determining the maximum injection pressure limit without initiating tensile and shear failures in the caprock. This will in turn impact the storage capacity of the reservoirs.

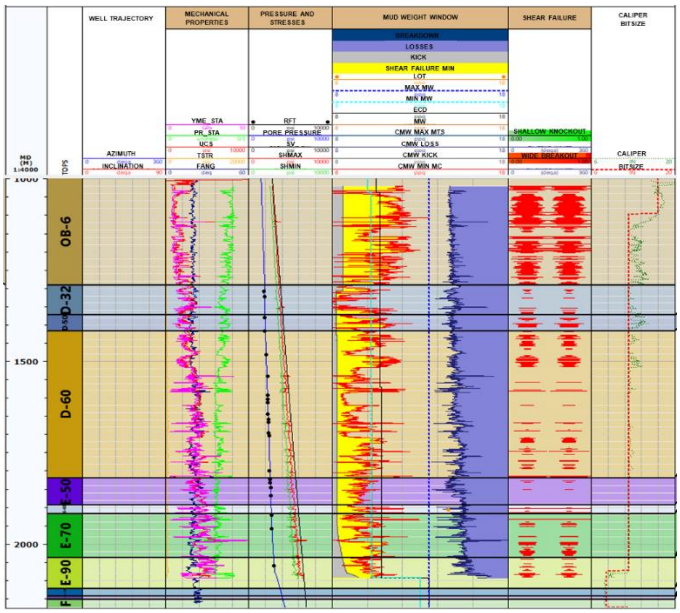


Fig. 3. 1D geomechanical model for an offset well and comparison of synthetic image log from wellbore stability analysis with hole enlargement based on caliper log.

When CO_2 is injected into the storage reservoir, it will result in increase in pore pressure and consequently, increase in total horizontal stresses in the reservoir. The relationship between the pore pressure and horizontal stress increase is given by Equation 1. It should be noted the equation can only be used for instantaneous changes of pore pressure and not to calculate the relationship between pore pressure and the horizontal stresses that are developed over geological time (Anderson, 1951).

$$\Delta\sigma_h = \alpha \frac{1-2v}{1-v} \Delta P_p \quad (1)$$

where:

$\Delta\sigma_h$ = Horizontal stress change
ΔP_p = Pore pressure change

The total horizontal stress change in the reservoir will cause reverse stress change of smaller magnitude in the immediate caprock (Tan et al., 2022) i.e., total horizontal stress increase in reservoir during injection will give rise to total horizontal stress decrease in the caprock to maintain stress equilibrium as shown in Fig. 4.

An upper limit horizontal stress transfer (depletion) and de-transfer (injection) ratio of 0.33 from reservoir to overburden due to production and injection was applied based on numerical modelling study. The recommended maximum injection pressure is based on the difference between the total horizontal stress of bottom of the caprock and the pressure at top of the reservoir of the wells. The recommended maximum injection pressures determined for the caprock seals at isothermal condition are summarised in Table 2. A safety factor of 1.5 was applied to the maximum delta injection pressure in view of the uncertainty of the input data.

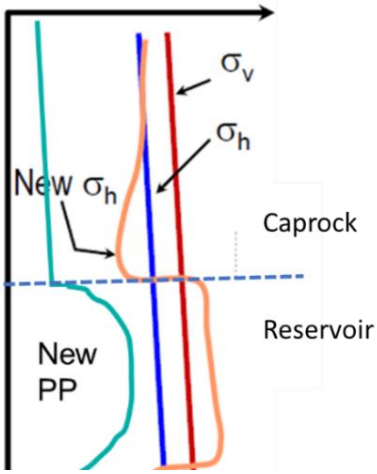


Fig. 4. Pore pressure and minimum horizontal stresses changes in reservoir and caprock.

Table 2. Recommended maximum injection pressure for caprock seals at isothermal condition.

Seal	Initial Reservoir Pressure Below Seal (psi)	Estimated Current Reservoir Pressure Below Seal (psia)	Original Caprock Total Shmin (psi)	Caprock Total Shmin After Depletion (psi)	Caprock Total Shmin After Injection psi)	Recommended Maximum Injection Pressure (psi)
Seal A	1847	450	3172	3373	3005	**2153**
Seal B	2050	1200	3728	3864	3496	**2731**
Seal C	2457	1100	4690	4936	4346	**3264**
Seal D	3015	1100	5334	5655	5001	**3701**

Thermal effect on the recommended maximum injection pressure was conducted analytically with an estimated 50 ^oC cooling in the caprock seals. There is considerable reduction in the injection pressure determined for the seals of up to 525 psi due to the reduction of the minimum horizontal stresses as summarised in Table 3. Similarly, a safety factor of 1.5 was applied to the maximum delta injection pressure in view of the uncertainty of the input data.

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Table 3. Recommended maximum injection pressure for caprock seals with CO_2 cooling.

Seal	Initial Reservoir Pressure Below Seal (psi)	Estimated Current Reservoir Pressure Below Seal (psia)	Original Caprock Total Shmin (psi)	Caprock Total Shmin After Depletion (psi)	Caprock Total Shmin After Injection psi)	Recommended Maximum Injection Pressure (psi)
Seal A	1847	450	3172	3373	2110	**1632**
Seal B	2050	1200	3728	3864	2713	**2209**
Seal C	2457	1100	4690	4936	3560	**2739**
Seal D	3015	1100	5334	5655	4217	**3178**

The caprock integrity analysis is based on shear failure analysis of the caprock. The Coulomb failure line is constructed from the rock cohesion and friction angle of the caprock at the depth of interest in solid black line. Sensitivity analysis of 0.9 of both cohesion and friction angle are plotted in dotted line. Mohr circles are plotted for the stress state at the four seals of interest above the injected reservoirs. The set of Mohr circle plots for Seal A are shown in Fig. 5 which correspond to prior to production, end of production and end of injection. The changes in the horizontal stresses and reduction in the effective stresses due to CO_2 injection will move the Mohr circle to the left side of the plots. If the Mohr circle touches the Coulomb failure line, the caprock will fail in shear failure mode. However, if the circle does not touch the Coulomb failure line and instead continues to move to the left side of the plot, it will fail in tensile failure mode. As the Mohr circles lie below the Coulomb failure line, it indicates no shear failure is expected throughout the three phases.

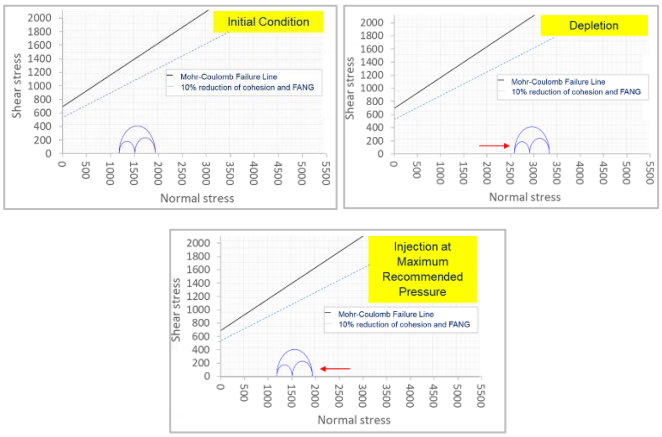


Fig. 5. Mohr circle plot for Seal A at initial, end of production and end of injection till initial reservoir pressure.

6.2. *Fault Stability Analysis*

After the recommended maximum injection pressures have been determined, fault stability analysis is conducted on the surrounding seismic interpreted faults in the field. From the total of 26 faults, the analysis focused on the major faults which are in proximity to the injected reservoirs within the region of interest. Fig. 6 shows the location of the analysed faults, and the rock properties and stresses contained in the 1D geomechanical models are used as input for the fault stability analysis.

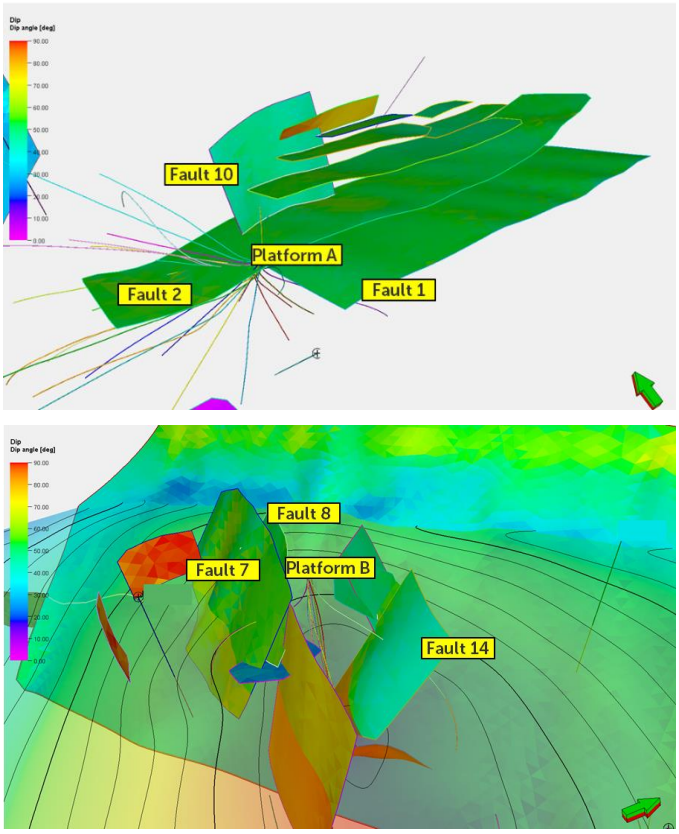


Fig. 6. Major faults near the vicinity of Platform A and Platform B.

Normal and shear stresses acting on the surface of Fault 1 are calculated and plotted in Fig. 7 for initial, end of production and end of injection phases. In this analysis, the fault is assumed to have zero cohesion, and the friction angle is based on the adjacent rock friction angle at depth of interest in the geomechanical model of the nearest offset well.

The following steps are conducted as part of fault stability sensitivity analysis:

- The fault friction angles used are reduced by 10% (solid line) and 20% (dotted line) from the friction angle of the adjacent rock.
- In the calculation of normal and shear stresses on the fault plane, the fault dip angle and fault azimuth are varied based on the distribution percentile (P10, P50 and P90).

If the analysis shows that the fault will re-activate at end of injection phase, the maximum injection pressure is reduced. The fault stress state lies below the Coulomb failure lines for the initial and end of injection phases. However, the 0.8 of adjacent rock friction angle dotted Coulomb failure line lies below a small number of fault stress state points which will result in localized failure but not fault re-activation. Hence, there is low risk for the fault to reactivate till end of injection and the maximum injection pressure need not be reduced.

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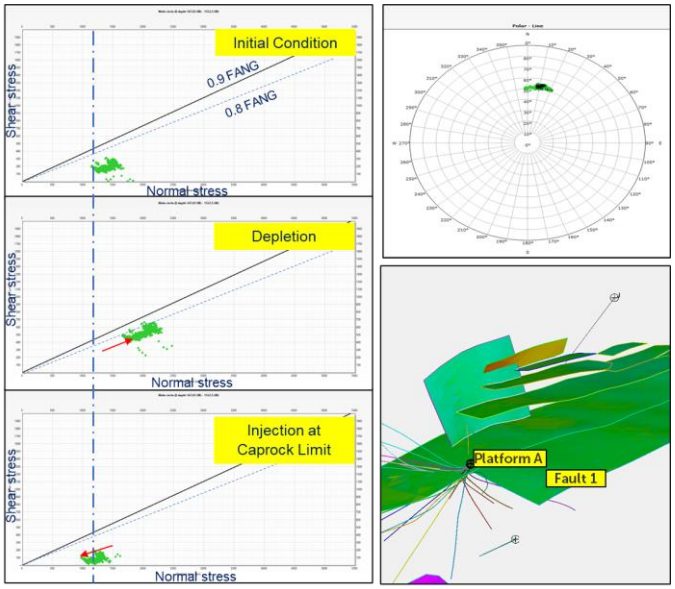


Fig. 7. Shear stress vs normal stress on Fault 1 surface at initial condition, end of production and end of injection (left); fault stereonet (top right); fault location with respect to Platform A and well (bottom right).

6.3. *Reservoir Expansion and Seabed Uplift Assessment*

With the maximum injection pressure being finalised, the seabed uplift due to the CO_2 injection is determined. When a reservoir is depleted, the pore pressure decline causes an increase in effective vertical stress, and this will result in the reservoir rock to deform and compact. The opposite holds true when a reservoir is injected. The pore pressure increase causes the reservoir to expand which is given by Equation 2.

$$\Delta h = \sum C_m \cdot \Delta P_p \cdot \Delta H \quad (2)$$

where:

Δh = Reservoir expansion
C_m = Coefficient of poroelastic uniaxial expansion
ΔH = Reservoir thickness

The coefficient of poroelastic uniaxial compaction/expansion in Equation 2 is defined in Equation 3.

$$C_m = \frac{1-\nu-2\nu^2}{(1-\nu)\mathrm{E}_{sta}} \quad (3)$$

where:

ν = Static Poisson's ratio
E_{sta} = Static Young's modulus

The C_m is calculated from the static elastic properties of the reservoir and the cumulative distribution of the coefficient shown in Fig. 8 was used to determine the P90, P50 and P10 values shown below:

Parameter	P90	P50	P10
Cm (1/MPa)	2.1E-05	4.7E-05	6.57E-05

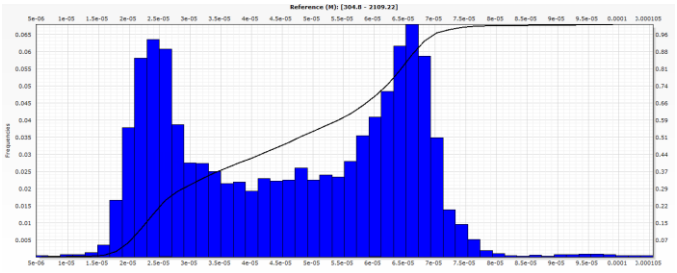


Fig. 8. Cumulative distribution of calculated C_m for reservoir.

The reservoir compaction and expansion are determined using Equation 2 and subsequently used for seabed subsidence and uplift calculations.

The seabed subsidence and uplift are calculated using two methods i.e., Geertsma Method (1973) and Bruno Method (1990). Calculation of the resultant seabed subsidence and uplift are based on Equation 4 and Equation 5 for Geertsma Method and Bruno Method, respectively.

Geertsma Method

$$S = -2(1-\nu)A\Delta h \quad (4)$$

where:

S = Seabed subsidence (uplift)
ν = Static Poisson's ratio
A = Geertsma integral value
Δh = Maximum compaction/expansion at top reservoir

Bruno Method

$$S = 2(1-\nu)[\Delta H - (R^2 + (D+\Delta H)^2)^{0.5} + (R^2 + D^2)^{0.5}]\left(\frac{\Delta H}{H}\right) \quad (5)$$

where:

ν = Static Poisson's ratio
ΔH = Maximum compaction/expansion at top reservoir
R = Average reservoir radius
D = Overburden thickness
H = Reservoir thickness

The resultant reservoir compaction and expansion and seabed subsidence and uplift for Low (P90), Base (P50) and High (P10) cases without rock properties degradation are shown in Table 3. The seabed uplift values are compared between the two methods and the highest value of 0.18 m is adopted.

The effects of rock properties degradation were analysed as part of the sensitivity analysis on reservoir compaction and expansion, and seabed subsidence and uplift for Low (P90), Base (P50) and High (P10) cases. Maximum reduction of 26% in Young's modulus and maximum increase of 10% in Poisson's ratio are applied in the analysis. There are significant differences of up to 37% decrease in the predictions as shown in Table 4.

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Table 3. Reservoir compaction and expansion, and seabed subsidence and uplift predictions for Low (P90), Base (P50), and High (P10) cases without rock properties degradation.

Stage	Process	Geertsma Method		
		P90	P50	P10
Depletion	Reservoir Compaction (m)	0.42	0.21	0.050
	Seabed Subsidence (m)	0.39	0.18	0.043
Injection	Reservoir Expansion (m)	0.19	0.078	0.016
	Seabed Uplift (m)	0.19	0.076	0.014

Stage	Process	Bruno Method			Maximum (m)
		P90	P50	P10	
Depletion	Reservoir Compaction (m)	0.42	0.21	0.050	0.42
	Seabed Subsidence (m)	0.38	0.20	0.043	0.39
Injection	Reservoir Expansion (m)	0.19	0.078	0.016	0.19
	Seabed Uplift (m)	0.16	0.075	0.014	0.19

Table 4. Reservoir compaction and expansion, and seabed subsidence and uplift predictions for Low (P90), Base (P50), and High (P10) cases with rock properties degradation.

Stage	Process	Geertsma Method		
		P90	P50	P10
Depletion	Reservoir Compaction (m)	0.42	0.21	0.050
	Seabed Subsidence (m)	0.39	0.18	0.043
Injection	Reservoir Expansion (m)	0.14	0.086	0.021
	Seabed Uplift (m)	0.12	0.076	0.020

Stage	Process	Bruno Method			Maximum (m)
		P90	P50	P10	
Depletion	Reservoir Compaction (m)	0.42	0.21	0.050	0.42
	Seabed Subsidence (m)	0.38	0.20	0.043	0.39
Injection	Reservoir Expansion (m)	0.14	0.086	0.021	0.14
	Seabed Uplift (m)	0.11	0.077	0.020	0.12

7. CONCLUSIONS

An analytical approach and workflow for assessing and mitigating CO_2 leakage risks associated with CO_2 injection for geological storage are described. They are illustrated via a preliminary study conducted on Field X which has been selected as a potential CO_2 storage site in offshore Peninsular Malaysia. The objective of the study is to evaluate the feasibility of injecting and storing CO_2 in the field by assessing CO_2 containment and leakage risks, in particular "showstoppers", for decision-making on progressing to comprehensive coupled geomechanical study and mature the field as a CO_2 storage site.

The results were subsequently used to develop recommendations from geomechanical perspective for CO_2 injection and storage operation in the potential storage site. As there is no "showstopping" geomechanical containment risks and the storage capacity and injectivity are favourable for CO2 storage site, the field is currently undergoing full field storage development plan study incorporating 3D/4D coupled geomechanical and thermal modelling which include rock property degradation due to CO_2-rock interaction. The information and workflow described may be adopted for preliminary feasibility evaluation of other CO_2 geological storage projects in both depleted and green carbonate and clastic reservoirs as well as saline aquifers worldwide.

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